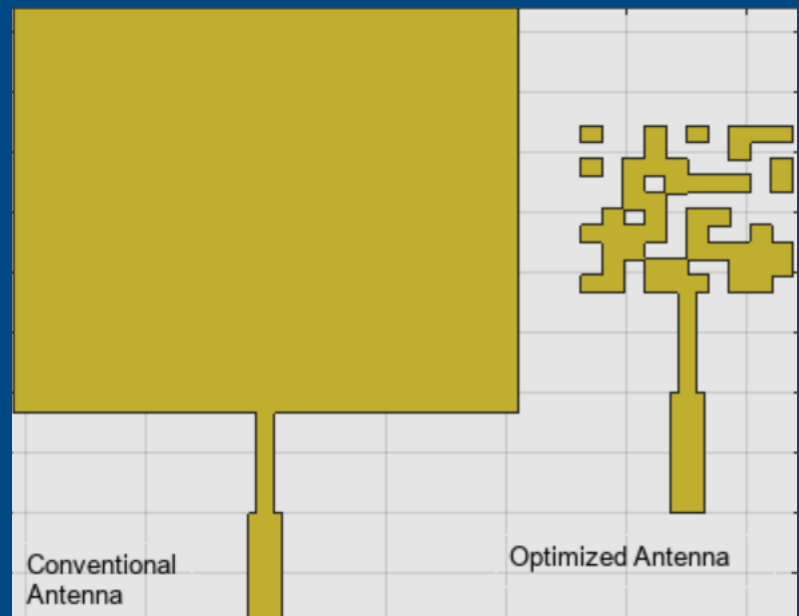
MILCOM 2027 — Tutorial TUT13
AI-Enabled Antenna-to-System Design with LLMs, MCP and MATLAB

Bonus — Antenna Optimization

Sekhar Sekharan– sekhars@mathworks.com

Applications Engineer
MathWorks | RF/EM Group

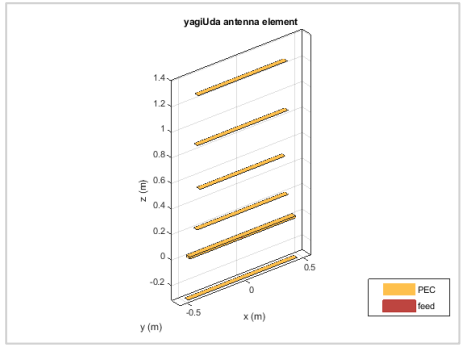
© 2026 The MathWorks, Inc.



Why optimize antenna designs?

Antenna design is a coupled trade-space problem, not a one-knob tuning problem.

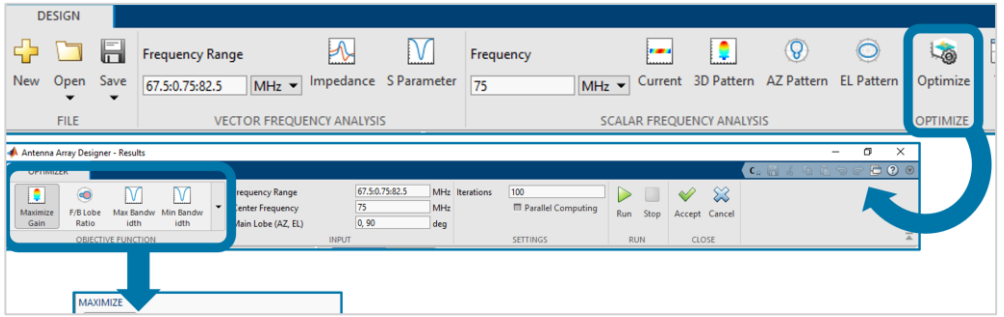
- Geometry variables interact: length, spacing, substrate, feed, array layout
- Targets compete: gain/directivity, return loss, bandwidth, side lobes, area
- Constraints matter: packaging, fabrication rules, and installed performance
- Optimization makes the search repeatable: variables + bounds + objective + constraints



Parameterized antenna geometry

		initialdesignVal	optimdesignVal
1	Reflector Length	0.9085	0.9361
2	Director Length - 1	0.9085	0.7836
3	Director Length - 2	0.9085	0.7318
4	Director Length - 3	0.9085	0.7799
5	Director Length - 4	0.9085	0.7638
6	Reflector Spacing	0.5451	0.3166
7	Director Spacing - 1	0.4542	0.2054
8	Director Spacing - 2	0.4542	0.3085
9	Director Spacing - 3	0.4542	0.3501
10	Director Spacing - 4	0.4542	0.3888

Candidate variables and optimized values



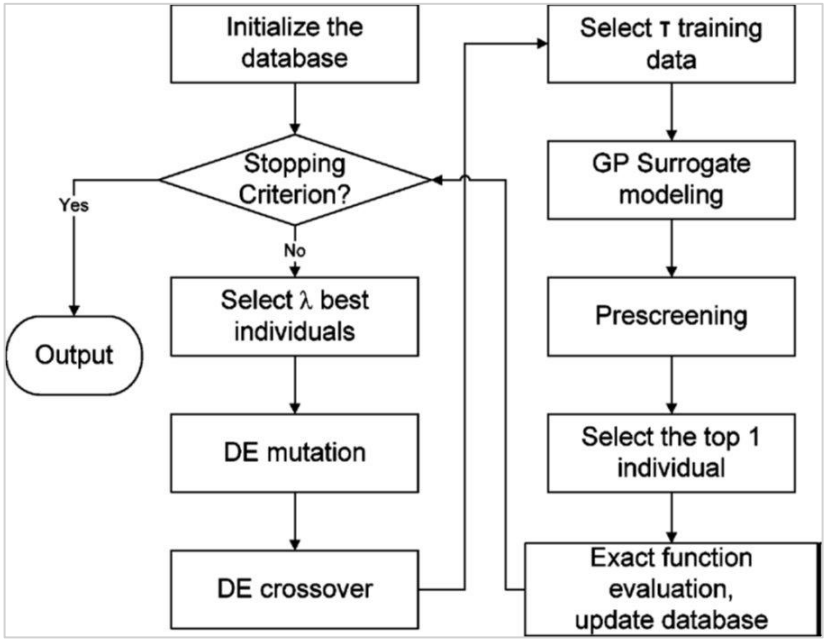
App workflow: choose objective, run, review, accept

Optimization converts manual antenna tuning into a measurable design search.

Why surrogate optimization?

Full-wave EM gives the truth, but evaluating every candidate design can dominate runtime.

- Direct global search can need many EM simulations
- Many proposed designs are poor and do not deserve a full solve
- A surrogate model learns from sampled EM results and prescreens new candidates
- Use it when design variables, constraints, or third-party EM loops make brute-force optimization too slow



Surrogate-assisted loop: sample → model → prescreen → EM verify

Keep EM as the source of truth; use the surrogate to decide where to spend EM time.

Surrogate-Based Optimization

Collaboration with Prof. Bo Liu, Univ. Glasgow

Surrogate model **A**ssisted **D**ifferential **E**volution for **A**ntenna synthesis (SADEA) is an AI-driven antenna design method

Initialization

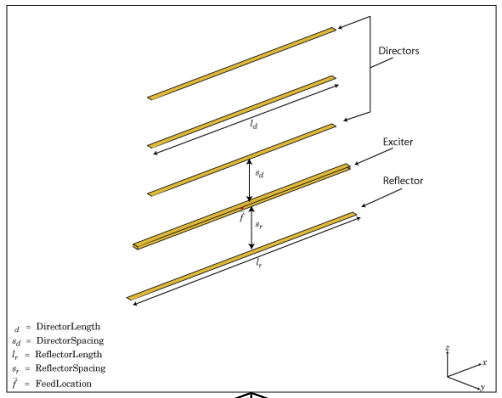
Generate α design samples from $[a\ b]^d$

Iteration

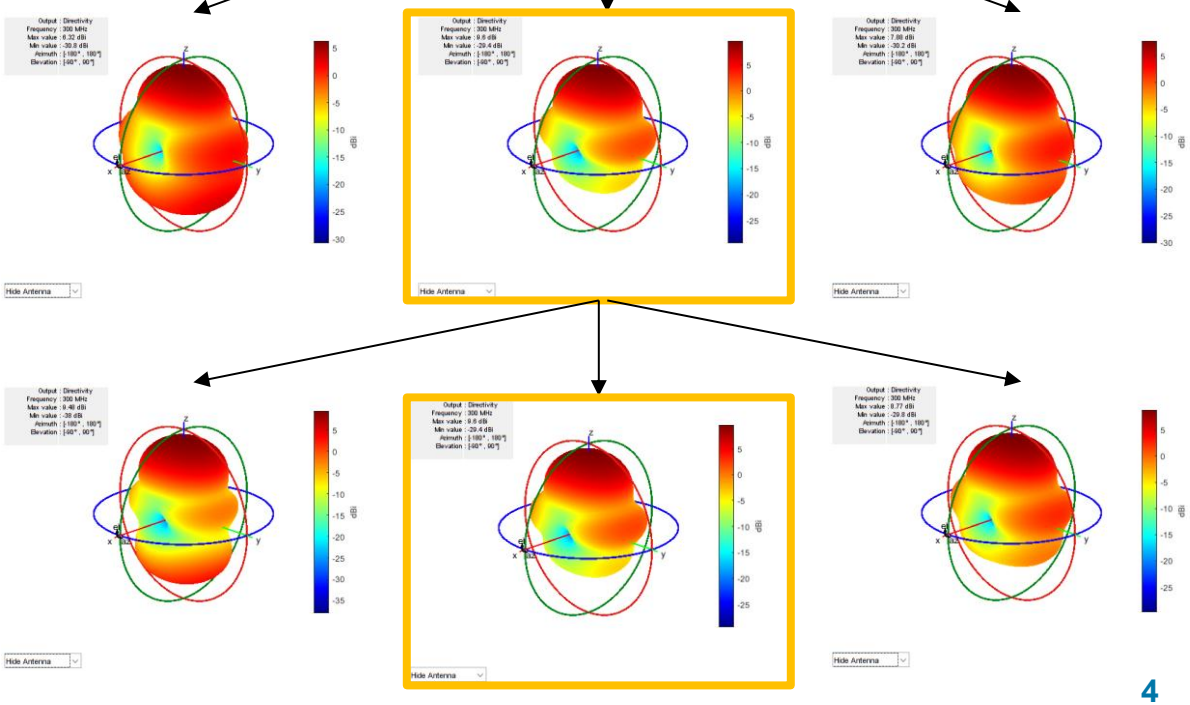
1. Select the β best candidate designs
2. Generate β child solutions
3. Prescreen β child solutions using Gaussian process surrogate model

Stopping criteria

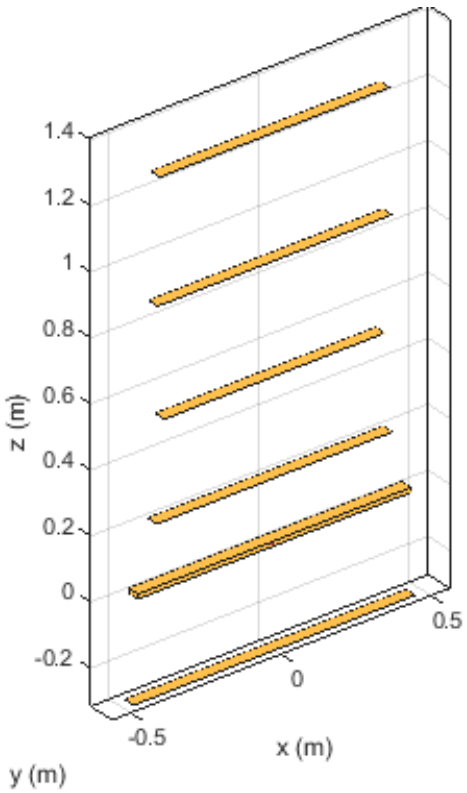
- The specifications are met
- Objective values does not improve for a certain number of iterations



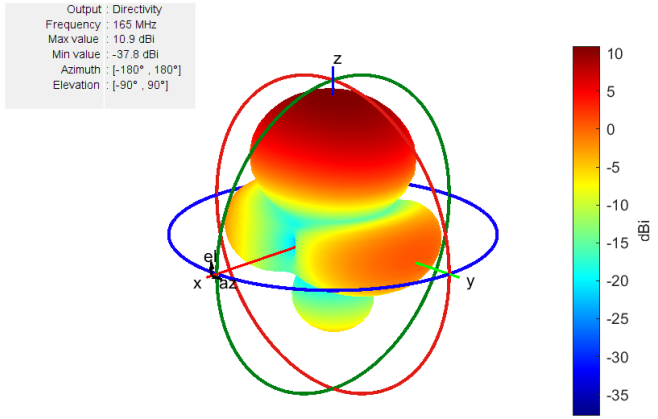
α design for reflector length from [0.1 1] meter



SADEA: Yagi-Uda Optimization Example

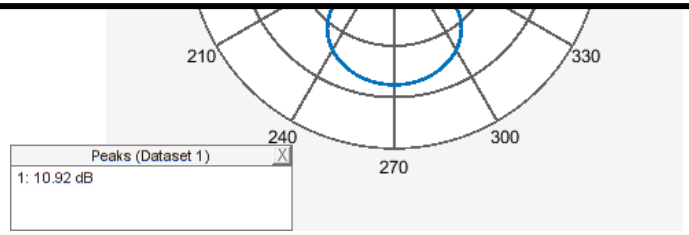


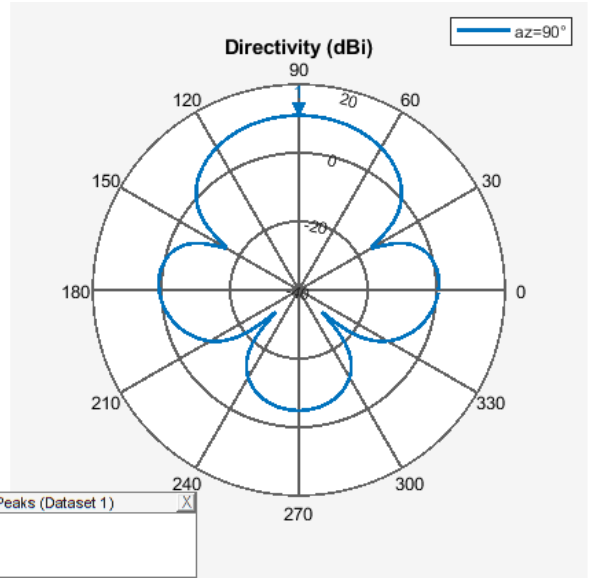
		initialdesignVal	optimdesignVal
1	Reflector Length	0.9085	0.9361
2	Director Length - 1	0.9085	0.7836
3	Director Length - 2	0.9085	0.7318
4	Director Length - 3	0.9085	0.7799
5	Director Length - 4	0.9085	0.7638
6	Reflector Spacing	0.5451	0.3166
7	Director Spacing - 1	0.4542	0.2054
8	Director Spacing - 2	0.4542	0.3085
9	Director Spacing - 3	0.4542	0.3501
10	Director Spacing - 4	0.4542	0.3888



	datasheetvals	optimdesignvals
Gain (dBi)	10.5	10.917
F/B	16	15.762
E-plane Beamwidth (deg.)	54	50
H-plane Beamwidth (deg.)	63	60
Impedance Bandwidth (%)	10	10

H-plane

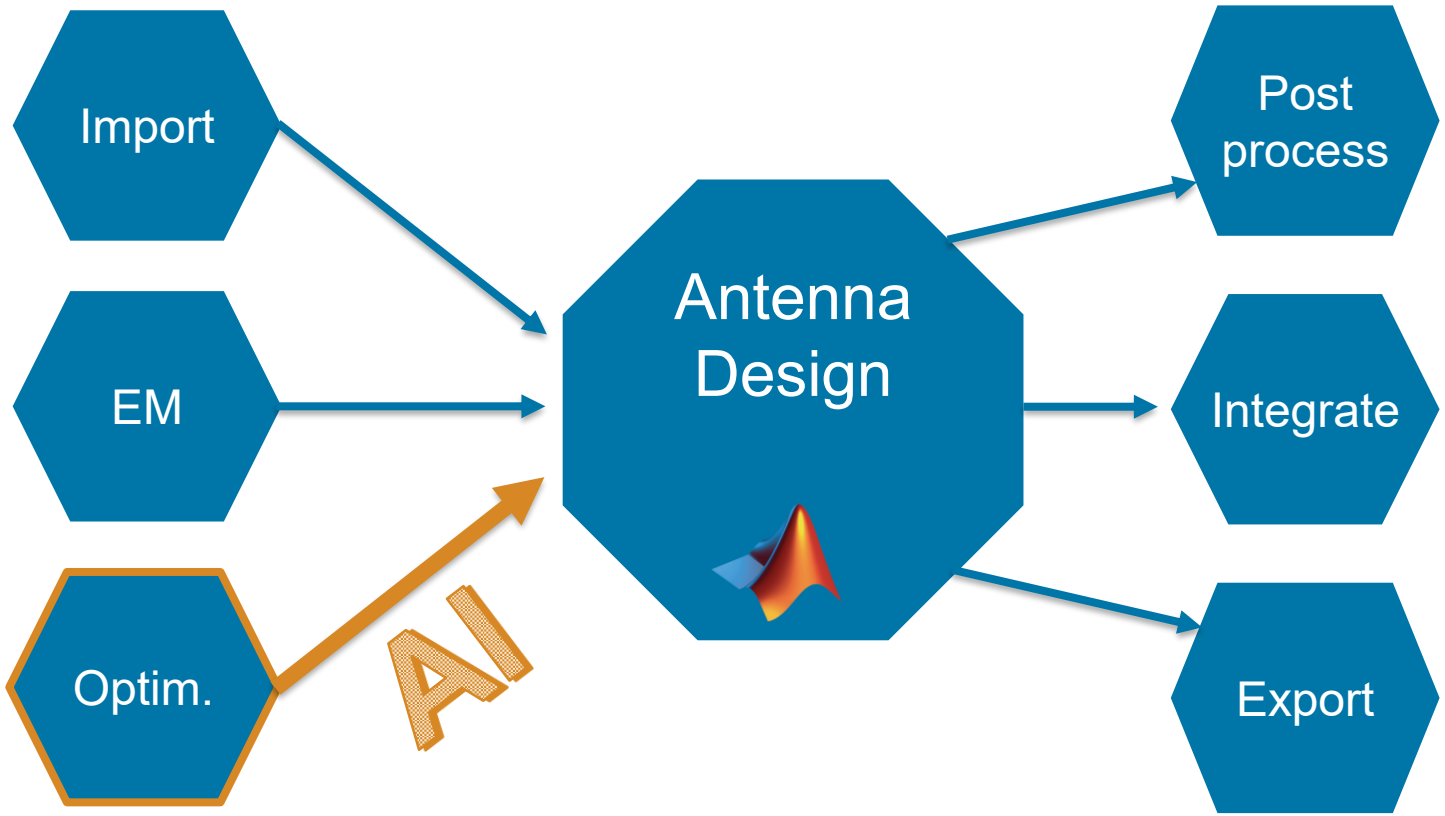




SADEA Optimization of Six-Element Yagi-Uda Antenna using Custom Objective Function
https://www.mathworks.com/help/antenna/ug/optimize-yagiuda-using-custom-objective-function.html

Optimization Exercise

Antenna Workflow #3: Optimization with (better) AI

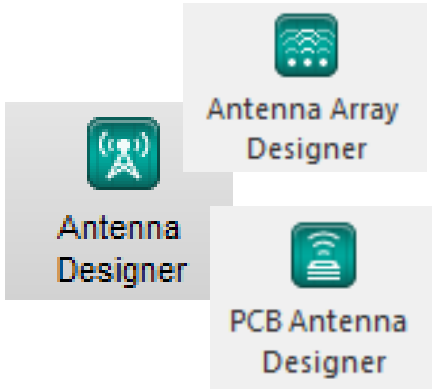


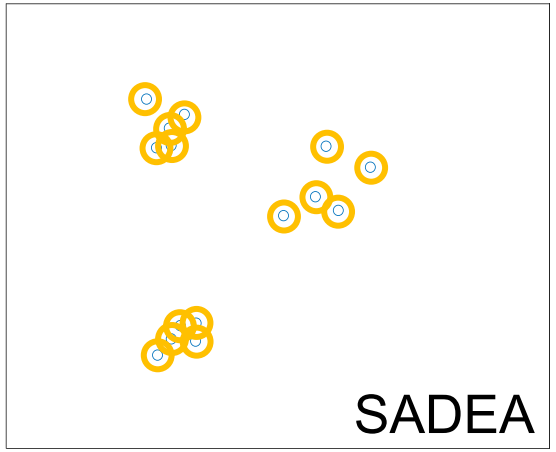
- Do you design antennas with a large number of design variables?
- Do you need to speed up antenna optimization?
- Do you use 3rd party EM tool and would like to use MATLAB for optimization?

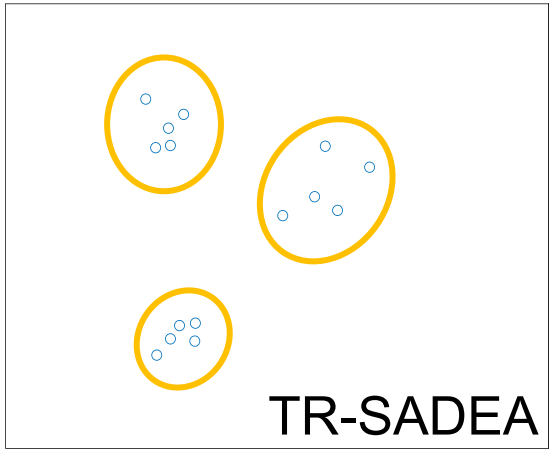
AI for Surrogate Based Optimization

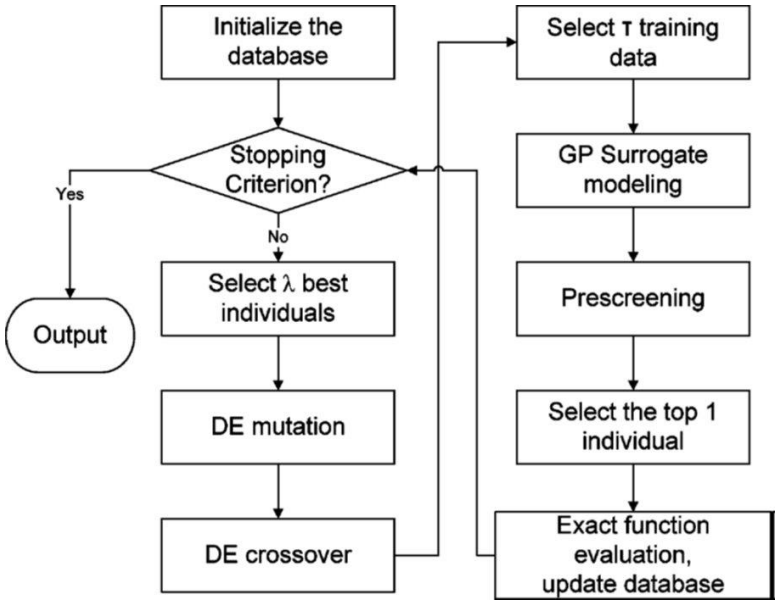
TR (Training-cost Reduced) – SADEA

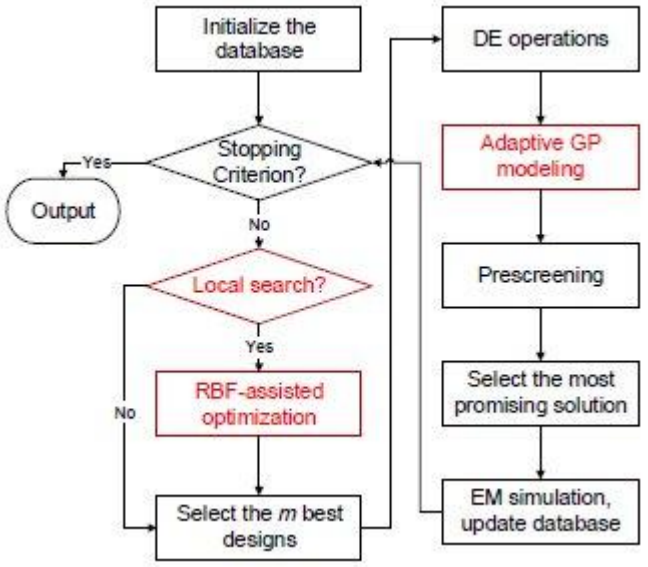
- Faster, while converging to a good solution
- Adaptive Gaussian Process predictions
- Handle up to 80 design variables
- Enabled in apps and command line



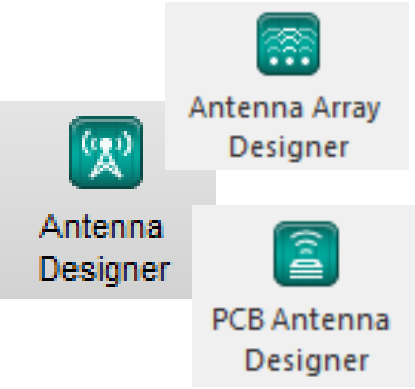




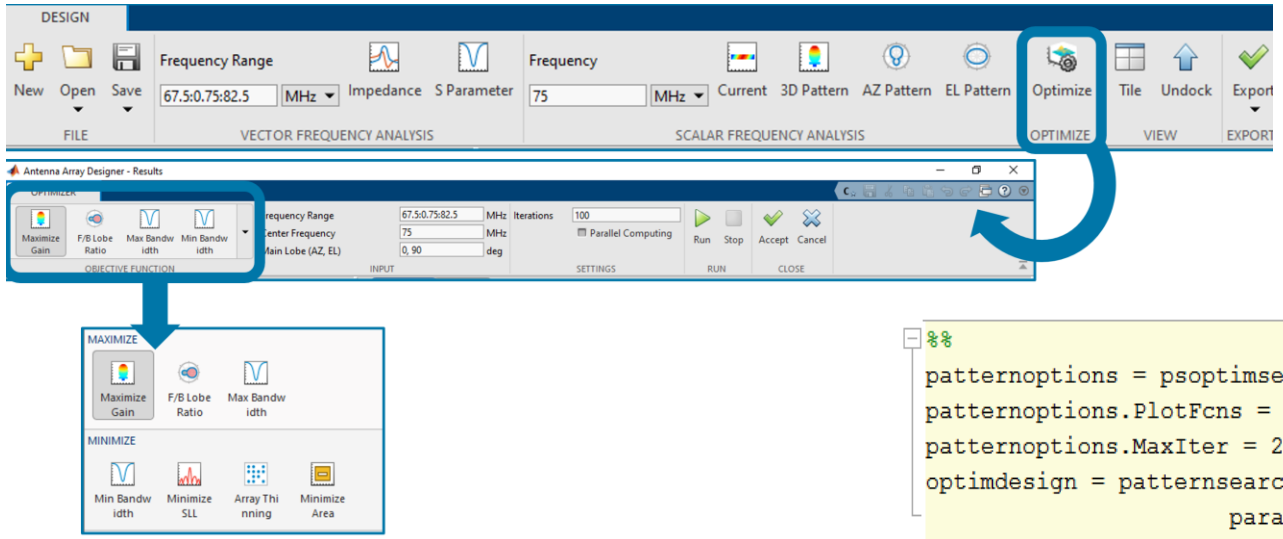




Antenna Optimization - Summary



- Standard objective functions available in the apps
  - SADEA, TR-SADEA, Surrogate optimization
  - Support for geometric constraints and integration with RF PCB Toolbox
- Custom objective functions can be defined at the command line
  - Interface for SADEA and TR-SADEA available
  - Can be used with 3rd party EM tools




```
%%
patternoptions = psoptimset(@patternsearch);
patternoptions.PlotFcns = @psplotbestf;
patternoptions.MaxIter = 25;
optimdesign = patternsearch(@(x) yagi_objective_function(yagidesign,x,freq,elang),...
                    parasitic_values,[],[],[],[],LB,UB,[],patternoptions);
```